

Mock Exam Round II

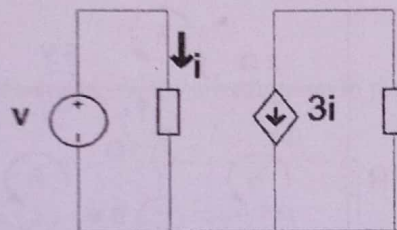
Core Courses

1. Which one the following statement is true regarding to the Ohm's law?

- A. All resistors not obey Ohm's law
- B. Linear resistor does not obey Ohm's law
- C. Ohm's law say that the Voltage (v) across a resistor is directly proportional to the resistance of resistor (R)
- D. For variable resistor, its current-voltage characteristic is always linear

Answer A

2. Carefully look at the following circuit in figure below. Which of the following is **NOT** true on the arrangement of electrical circuit element.



- A. There is no independent current source in the circuit
- B. There is one dependent current source in the given circuit
- C. The dependent source is Current-Controlled voltage source
- D. There is current-controlled current source

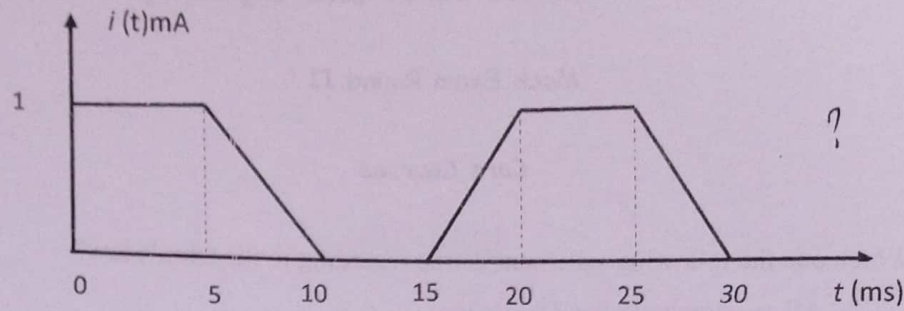
Answer C

3. The charge entering the positive terminal of an element is $q = 5 \sin(4\pi t)$ mC while the voltage across the element (plus to minus) is $v = 3\cos(4\pi t)$ V. What is the total energy delivered to the element between 0 and 6 s.

- A. 18J
- B. 18π J
- C. 17π J
- D. 5J

Answer B

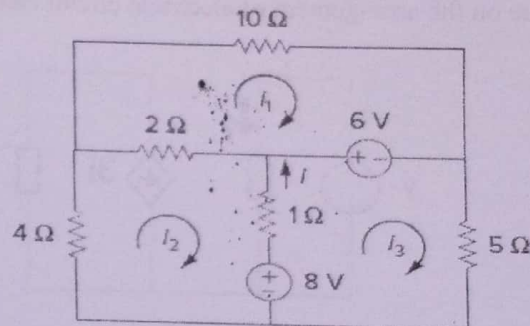
4. The current that enters an electrical circuit element is given in the Figure below. What is the charge that enters the element in the time interval of $0 < t < 30$ s is



- A. 0.105C
- B. 0.75C
- C. 0.2C
- D. 0.175C

Answer D

5. The direction of mesh current in the following circuit is given. What is the current (I) flowing through the 1Ω resistor



- A. 1.188A
- B. 2.25A
- C. 3.187A
- D. 1.8A

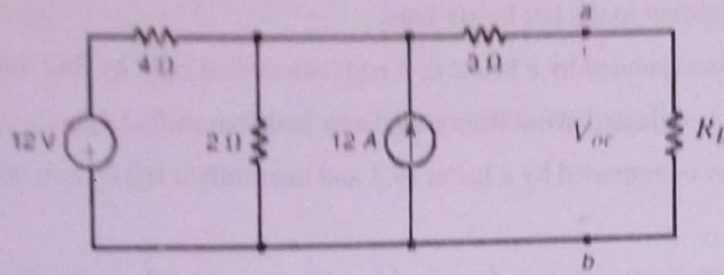
Answer A

6. Which of the following is true about fundamental circuit theorems (Thevenin and Norton theorem)

- A. Originally the theorem was developed to analysis the complex circuit such as the circuit contain active element
- B. The theorems are applicable for the circuit which contain passive and active components but they must be linear circuit
- C. The theorems are applicable only to the linear resistive and reactive circuits
- D. Power dissipation for the Thevenin circuit is always the same as that of the real system

Answer C

7. What is the open circuit voltage V_{oc} at terminals a and b of the circuit in the figure



- A. 8V
- B. 12V
- C. 4V
- D. 10V

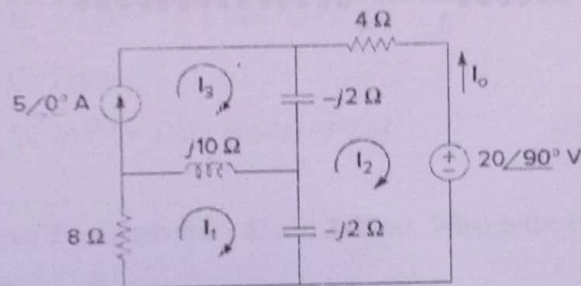
Answer B

8. Which of the following is **Not** true regarding to the impedance of the circuit

- A. Impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I
- B. Impedance represents the opposition that the circuit exhibits to the flow of sinusoidal current
- C. Impedance is sinusoidally varying quantity
- D. Impedance has both real and imaginary parts

Answer C

9. The Mesh current I_2 in the circuit given in the figure below is__



- A. $6.12 \angle -35.22^\circ \text{ A}$
- B. $6.12 \angle 144.78^\circ \text{ A}$
- C. $416.17 \angle -35.22^\circ \text{ A}$
- D. $5.969 \angle 65.45^\circ \text{ A}$

Answer A

10. Amongst the following expression which one is false?

- A. $t^2 \delta(t) = 0$
- B. $\sin(t) \delta(t - \pi) = \delta(t - \pi)$
- C. $\cos(t) \delta(t) = \delta(t)$
- D. $\int_{-\infty}^{\infty} (2t^2 + 1) \delta(t) dt = 1$

Answer B

11. Given the signal $x(t)$, which of the following is true for signal $x(3t + 6)$?

- A. $x(t)$ shifted to the left by six units
- B. $x(t)$ is expanded by a factor of 3 and then shifted right by four units
- C. $x(t)$ is reflected about the vertical axis and then shifted 4.
- D. $x(t)$ is compressed by a factor of 3 and then shifted left by two units

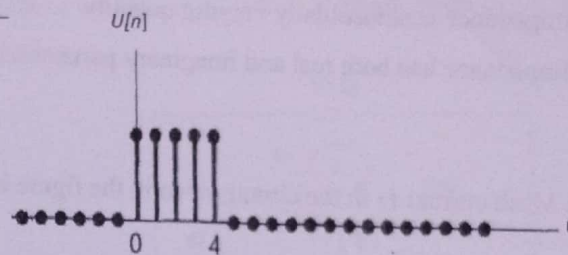
Answer D

12. Continuous time system is described by $y(t) = \cos x(t) + 3\sin x(t - 2)$. which one of the following is correct about $y(t)$

- A. Causal and with memory
- B. Noncausal and with memory ?
- C. Only Memoryless
- D. with memory and non-causal

Answer A

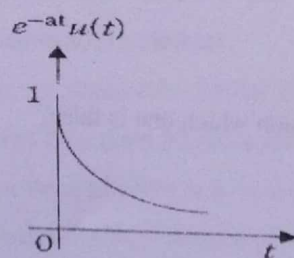
13. The equivalent mathematical expression of the discrete time unit step signal in figure below is _____



- A. $u[n] - 4u[n]$
- B. $u[n] - u[n-4]$
- C. $u[n] - u[n+4]$
- D. $u[n] - r[n-4]$

Answer B

14. What is the Laplace transform of the signal in Figure shown below?



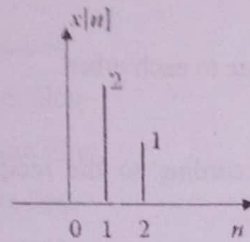
- A. $1/s - a$
- B. $1/s$

C. 0

D. $1/s+a$

Answer D

15. The z-transform of the sequence shown in Figure below is _____



A. $2z + z^2$

B. $2z^{-1} + z^{-1}$

C. $2z^{-1} + z^{-2}$

D. $z^{-1} + 2z^{-2}$

Answer C

16. Given the Signal $x(t) = U(3t) + \delta(t-2)$. The Fourier transform $X(\omega)$ which transform the signal from time domain to frequency domain can be computed as

A. $\pi\delta(\omega) + 1/j\omega$

B. $1/3 [e^{-j2\omega} + \pi\delta(\omega/3) + 1/j\omega]$

C. $2\pi\delta(\omega)$

D. $e^{-j2\omega} + 1/3[\pi\delta(\omega/3) + 1/j\omega]$

Answer D

17. Given the signals $\delta(t-3)$ and $2\sin\omega_0 t$. What is the Fourier transform of this signal respectively

A. 1 and $4\pi\delta(\omega_0)$

B. $e^{-j3\omega}$ and $2\pi j[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

C. 1 and $2\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$

D. $e^{j3\omega}$ and $2\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

Answer B

18. When the excitation is a current source and the response is a voltage then the system function is _____

A. Admittance

B. Impedance

C. Voltage-ratio transfer function

D. Current-ratio transfer function

19. A network function is said to have simple pole or simple zero if

- A. The poles and zeroes are on the $j\omega$ axis
- B. The poles and zeroes are on the real axis
- C. The poles and zeros are not repeated
- D. The poles and zeroes are complex conjugate to each other

Answer C

20. If the given network is reciprocal, then according to the reciprocity theorem h-parameters

- A. $h_{12} + h_{21} = 0$
- B. $h_{12} - h_{21} = 0$
- C. $h_{12}h_{21} = h_{11}h_{22}$
- D. B and C are the answer

Answer A

21. Which of the following is TRUE regarding a terminated two-port network?

- A. For an open circuited output $Z_L \rightarrow \infty, Z_{in} = z_{11}$
- B. For a short-circuited output $Z_L = 0, Z_{in} = \frac{\Delta z}{z_{22}} = \frac{1}{y_{11}}$
- C. For an open circuited input $Z_L \rightarrow \infty, Z_{in} = z_{22}$
- D. All of the above are true

Answer D

22. Transient phenomena occur whenever

- A. There is a sudden change in the applied voltage from one finite value to another
- B. A circuit is suddenly connected or disconnected from the supply
- C. A circuit is short-circuited
- D. All

Answer D

23. A Hurwitz polynomial has

- A. Zeros only in the left half of the s-plane
- B. Poles only in the left half of the s-plane
- C. Zeros anywhere in the s-plane
- D. Poles on the $j\omega$ axis only

Answer B

24. If a filter is de-normalized to a higher frequency, which of the following occurs

- A. Inductors increase in value while capacitors decrease
- B. Inductors and capacitors increase in value

- C. Inductors and capacitors decrease in value
- D. Inductors decrease in value while capacitors increase

Answer D

25. The voltage ratio transfer function of an active filter is given by $\frac{V_2(s)}{V_1(s)} = \frac{s^2 + \delta}{s^2 + \alpha s + \delta}$. The

circuit question is _____

- A. Band pass filter
- B. Band reject filter
- C. High pass filter
- D. Low pass filter

Answer B

26. The time measurement between the 90% point on the trailing edge of a pulse to the 10% point on the trailing edge of the pulse is called the _____

- A. rise time
- B. fall time
- C. period
- D. pulse width

Answer B

27. Assume that a floating-point number is represented in binary. If the sign bit is 1, the

- A. number is negative
- B. number is positive
- C. exponent is negative
- D. exponent is positive

Answer A

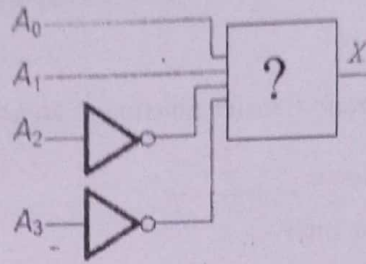
28. To implement the expression the $X = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}D + B\bar{D}E$ type of gate that is needed is

- A. 3-input AND gate from OR gates
- B. 3-input NAND gate from NOR gates
- C. 3-input OR gate from AND gates
- D. 3-input NOR gate from NAND gates

Answer C

29. Assume you want to decode the binary number 0011 with an active-HIGH decoder. The missing gate should be

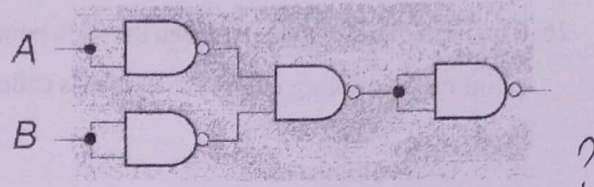
- A. an AND gate
- B. an OR gate
- C. a NAND gate
- D. a NOR gate



Answer A

30. The circuit shown is equivalent to an

- A. AND gate
- B. XOR gate
- C. OR gate
- D. none of the above



Answer C

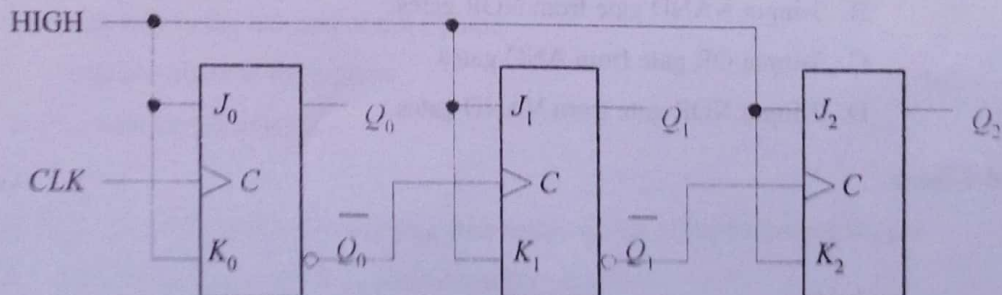
31. How many AND gates are required to construct a 4 – bit parallel multiplier if four 4 – bit parallel binary adders are given?

- A. Four 2 – input AND gates
- B. Eight 2 – input AND gates
- C. Sixteen 2 – input AND gates
- D. Two 2 – input AND gates

Answer C

32. The Q_0 output of the counter shown

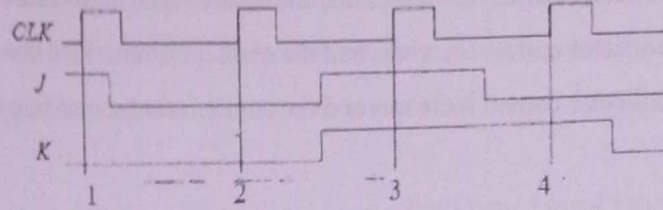
- A. is present before Q_1 or Q_2
- B. has a higher frequency than Q_1 or Q_2
- C. changes on every clock pulse
- D. all of the above



Answer D

33. Assume the output of the following PET J-K flip-flop is initially HIGH. On which clock pulse will the output of the flip-flop go from HIGH to LOW?

- A. 4
- B. 3
- C. 2
- D. 1



Answer B

34. The depletion region of a semiconductor diode is due to

- A. reverse biasing
- B. crystal doping
- C. forward biasing
- D. migration of mobile charge carriers

Answer D

35. The process of converting ac to pulsating dc is called

- A. clipping
- B. charging
- C. rectification
- D. filtering

Answer C

36. The output frequency of a full-wave rectifier with a 60 Hz sinusoidal input is

- A. 30 Hz
- B. 60 Hz
- C. 120 Hz
- D. 0 Hz

Answer C

37. During the positive half-cycle of the input voltage in a bridge rectifier,

- A. one diode is forward-biased
- B. all diodes are forward-biased
- C. all diodes are reverse-biased
- D. two diodes are forward-biased

Answer D

38. If the base current of a Bipolar junction transistor operating in the linear region increases,

- A. the collector current increases and the emitter current decreases
- B. the collector current decreases and the emitter current decreases
- C. the collector current increases and the emitter current does not change
- D. the collector current increases and the emitter current increases

Answer D

39. When an n-channel JFET is biased for conduction, the gate is

- A. positive with respect to the source
- B. negative with respect to the source
- C. positive with respect to the drain
- D. at the same voltage as the drain

Answer B

40. In a class-A amplifier, conduction extends over 360° because Q-point is

- A. located on load line
- B. located near saturation point
- C. centered on load line
- D. located at or near cut-off point

Answer C

41. Cascading two amplifiers will result in

- A. reduction in overall gain and increase in overall bandwidth.
- B. reduction in overall gain and reduction in overall bandwidth
- C. increase in overall gain increase in overall bandwidth
- D. increase in overall gain and reduction in overall bandwidth

Answer C

42. The amplifier in which the emitter and collector leads of one transistor are connected to the base and collector leads of a second transistor is .

- A. push-pull amplifier
- B. Darlington amplifier
- C. differential amplifier
- D. complementary amplifier

Answer B

43. What is the difference between struct and class in C++?

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- A. All members of a structure are public and structures don't have constructors and destructors
- B. Members of a class are private by default and members of struct are public by default. When deriving a struct from a class/struct, default access-specifier for a base class/struct is public and when deriving a class, default access specifier is private
- C. All members of a structure are public and structures don't have virtual functions
- D. All of the above

Answer B

44. What is the return value of f(p, p) if the value of p is initialized to 5 before the call?
Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f(int &x, int c)
{
    c = c - 1;
    if (c == 0) return 1;
    x = x + 1;
    return f(x, c) * x;
}
```

- A. 3024
- B. 6561
- C. 55440
- D. 161051

Answer B

45. How can we restrict dynamic allocation of objects of a class using new?
- A. By making an empty private new operator
 - B. By overloading new operator
 - C. By making an empty private new and new[] operators
 - D. By overloading new operator and new[] operators

Answer C

46. Which of the following is true about this pointer?
- A. It is passed as a hidden argument to all function calls
 - B. It is passed as a hidden argument to all non-static function calls
 - C. It is passed as a hidden argument to all static functions
 - D. None of the above

B

47. What is the output for the following program?

```
#include<iostream>
using namespace std;
int fun(int x = 0, int y = 0, int z)
{ return (x + y + z); }
int main()
{
    cout << fun(10);
    return 0;
}
```

A. 10

☒ B. 0

C. 20

D. Compiler Error

Answer D

48. What is the output for the following program?

```
#include<iostream>
using namespace std;
int x = 1;
void fun()
{
    int x = 2;
    {
        int x = 3;
        cout << ::x << endl;
    }
}
int main()
{
    fun();
    return 0;
}
```

A. 1

B. 2

C. 3

D. 0

Answer A

49. What is the value of p in the following C++ code snippet?

```
#include <iostream>
```

```
using namespace std;
int main()
{
    int p;
    bool a = true;
    bool b = false;
    int x = 10;
    int y = 5;
    p = ((x | y) + (a + b));
    cout << p;
    return 0;
}
```

A. 12

B. 0

C. 2

D. 16

Answer D

50. Which concept allows you to reuse the written code in C++?

A. Inheritance

B. Abstraction

C. Polymorphism

D. Encapsulation

Answer A